

Day 23 — Transcript Data Architecture

Objective: Define how voice conversations are converted into structured, AI-processable data.

Day 23 Tasks

Task	Status
Design transcript storage structure	Completed
Define metadata standards	Completed
Define transcript normalization rules	Completed
Design database schema for screening interactions	Completed

1. Voice Transcript Schema

A structured Python dataclass was created in `interview_ai/transcript_schema.py` to represent one voice transcript.

```
@dataclass
class Transcript:
    transcript_id: str
    candidate_id: str
    job_id: str
    question_id: str
    speaker: str
    transcript_text: str
    timestamp: datetime
    confidence: float
```

Test Result

The schema was tested successfully by creating a transcript object with candidate, job, question, timestamp, text, and confidence information.

2. Metadata Standards

Field	Standard	Example
candidate_id	CAND-<number>	CAND-001
job_id	JOB-<role>-<number>	JOB-AI-001
question_id	Q-<number>	Q-001
timestamp	ISO 8601 datetime	2026-08-18T17:00:00
confidence	Decimal from 0.0 to 1.0	0.96

The metadata standards ensure transcript records use consistent identifiers, timestamps, and speech-to-text confidence values.

3. Transcript Normalization

A normalization function was created in `interview_ai/transcript_normalizer.py`. It removes common filler words, leftover filler punctuation, repeated spaces, and spaces before punctuation.

```
Raw:
uhh... I have    two years um of experience in Python .
```

```
Normalized:  
I have two years of experience in Python.
```

The normalization test passed successfully.

4. Screening Interaction Schema

A structured screening interaction model was created in `interview_ai/screening_interaction.py`.

```
@dataclass  
class ScreeningInteraction:  
    interaction_id: str  
    candidate_id: str  
    job_id: str  
    question_id: str  
    transcript_id: str  
    answer_text: str  
    timestamp: datetime  
    confidence: float
```

The schema was tested successfully with a sample screening interaction linked to transcript TRN-001.

Project Files Created

```
interview_ai/  
■■■ transcript_schema.py  
■■■ transcript_normalizer.py  
■■■ screening_interaction.py
```

Day 23 Completion

Day 23 core implementation completed. The transcript schema, metadata standards, normalization rules, and screening interaction structure were implemented and tested.